Project Report - Day 7

Task-07: Building a Vector-Based Search Engine using Sentence Embeddings

Objective

The objective of this task is to build a simple semantic search system that retrieves the most relevant sentences from a text corpus based on the user's query. The task focuses on converting both the query and corpus sentences into embeddings, comparing them using cosine similarity, and returning the top matching results.

Libraries Used

1. **pathlib**
   - Used to build a reliable file path for loading corpus.txt from the same folder as the script.
2. **sentence_transformers**
   - Used to load the pretrained model all-MiniLM-L6-v2 and generate embeddings for both corpus sentences and the user query.
3. **sklearn.metrics.pairwise**
   - Used for cosine_similarity to compare the query embedding with corpus embeddings.
4. **huggingface_hub**
   - Used to log in with the Hugging Face token when available.
5. **dotenv**
   - Used to load environment variables from the .env file.

Errors

1. **Model loading issue**
   - **Issue:** The sentence transformer model could fail to load if the model was not available locally or if access was restricted.
   - **Solution:** Optional Hugging Face login was added using HF_TOKEN so the model can be accessed when needed.
2. **Empty query input**
   - **Issue:** If the user entered an empty query, the program could not perform semantic search properly.
   - **Solution:** Input validation was added to check for empty input and display a proper message.
3. **File path issue**
   - **Issue:** The program might not find corpus.txt if it was run from a different working directory.
   - **Solution:** The script was updated to load the file using Path(**file**).resolve().parent so the correct path is always used.
4. **No relevant result found**
   - **Issue:** Sometimes the returned results were not sufficiently related to the entered query.
   - **Solution:** A similarity threshold of 0.5 was added so the script only shows results when they are relevant enough.
5. **Unnecessary warning messages**
   - **Issue:** Extra warnings from libraries made the output less clear.
   - **Solution:** warnings.filterwarnings("ignore") was added to suppress unwanted warnings and keep the output clean.

Output

```
PS D:\Daily_Tasks> & C:\Users\Acer\AppData\Local\Programs\Python\Python310\python.exe d:/Daily_Tasks/Day7_Task/semantic_search.py
Note: Environment variable`HF_TOKEN` is set and is the current active token independently from the token you've just configured.
Loading weights: 100%|| 103/103 [00:00<00:00, 1360.32it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Enter your search query: Data Scientist

Top Results:

Score: 0.5899999737739563
46.     Data scientists spend most of their time cleaning and preprocessing raw data.
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Score: 0.46999999880079071
9.      Python is the primary language used for data science and machine learning.
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Score: 0.38999998569488525
1.      Artificial intelligence is transforming the way we write code in VS Code.
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PS D:\Daily_Tasks> & C:\Users\Acer\AppData\Local\Programs\Python\Python310\python.exe d:/Daily_Tasks/Day7_Task/semantic_search.py
Note: Environment variable`HF_TOKEN` is set and is the current active token independently from the token you've just configured.
Loading weights: 100%|| 103/103 [00:00<00:00, 3994.65it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Enter your search query: Actor

Top Results:

No relevant results found in the file.
PS D:\Daily_Tasks>
```